Terminology mapping to the brief. The Datathon brief (§3) lists four impact-materiality factors: *scale, scope, likelihood, irreversibility.* Our formulas preserve Aqualia's published nomenclature. The mapping is one-to-one:

Brief term	Our term	Relationship
Scale	Scale	identical
Scope	Scope	identical
Likelihood	Probability	synonymous (P[occur] on the 1–5 scale)
Irreversibility	Remediability	inverse (high remediability implies low irreversibility)

The factor set is identical to the brief; the aggregation pattern follows Aqualia's published practice rather than a single 4-factor mean. The Monte Carlo pipeline propagates uncertainty in all four factors simultaneously, so no factor is suppressed in the final position.

3.3 Threshold calibration

Threshold bands for every sub-criterion are documented in the scoring rubric (Appendix A). Anchors include ESRS 1 §43–51, EFRAG IG-1 [5], SASB Water Utilities metrics [13] (non-revenue water, pipe replacement, effluent non-compliance), WRI Aqueduct population-exposure bands [11], and MSCI 2024 Water Utilities sub-weights [16].

The Financial Magnitude scale is anchored to Aqualia's 2024 revenue. Our base case uses €1.4 billion (Assumption A01) as a conservative proxy. Aqualia's 2024 Sustainability Report (p. 103) discloses total revenue-volume of business of **€1,674.7 M** [10], of which 98.8% is EU-Taxonomy-eligible and 45.74% Taxonomy-aligned. The conservative anchor sits inside our published ±20% sensitivity band; rank ordering and the sign of the €18 M/yr headline are stable under both anchors.

3.4 Stakeholder salience

Aqualia's 2025 review was scored internally without external stakeholder consultation, as their methodology acknowledges [15]. We restore a salience input by applying the Mitchell–Agle–Wood (1997) model [2], the canonical framework in stakeholder theory, to Aqualia's own ten-group stakeholder map. Each stakeholder is scored 0/1 on Power, Legitimacy, and Urgency; the salience score $\in \{0, 1, 2, 3\}$ becomes a normalised weight in $[0, 1]$ summing to 1.000. Four groups reach Definitive status (weight 0.120 each): Public authorities and regulators, Customers and end-users, Shareholders, and Employees. Six groups reach Expectant status (weight 0.080 each): Investors, Suppliers, Business partners, Society, Local communities, and Environment. Academia reaches Discretionary status (weight 0.040). The full table is in Appendix D.

3.5 Monte Carlo design

Every uncertain input feeds a triangular distribution centred on the base value with perturbation ±0.7 on the 1–5 scale, clipped to $[1, 5]$. Stakeholder weights sample from a Dirichlet distribution centred on the salience vector (concentration parameter $\kappa = 100$). Financial magnitude uses triangular (low, base, high) as specified in the assumption log (Appendix B). Ten thousand draws are generated per topic. For each topic, the (Financial, Impact) scatter produces a 2D distribution from which a 90% χ^2-containment confidence ellipse is drawn via eigendecomposition of the covariance matrix ($\chi^2_{2,\,0.90} \approx 4.605$; 1σ semi-axis scales by $\sqrt{4.605} \approx 2.146$).

3.6 Robustness

The Monte Carlo is repeated under four alternate stakeholder-weighting schemes: Equal (all eleven groups at 1/11); Regulator-heavy (Public Authorities at 0.30); Community-first (Society + Local Communities combined at 0.30); and Investor-first (Shareholders + Investors combined at 0.40). **No ranking flip occurs** across schemes: T1 and T2 remain in the Target Zone; T3 remains borderline. Stability constitutes our robustness claim. A sensitivity tornado per topic is in §7.2.

4. Topic Identification and Peer Benchmark

4.1 From sixteen topics to three

Aqualia's 2025 review [17] identifies sixteen topics and 83 IROs. Our task is to select three topics that: (i) reflect peer and regulator consensus on sector-core issues; (ii) surface Aqualia-specific differentiation; and (iii) act as an organising structure for a 2027–2030 strategy. Selection proceeds through sector-core overlay, blind-spot overlay, and differentiator overlay.

4.2 Peer benchmarking

We construct a fourteen-column benchmark comparing Aqualia's sixteen topics to the declared material topics of five peers (Veolia [18], Suez [19], Saur, Facsa, Global Omnium [20], the five named in Aqualia's own 2025 benchmark) and with Severn Trent [21] as a CSRD-adjacent UK comparator. The full matrix is in Appendix C.

Shared sector core (universal across peers).. Climate mitigation and adaptation (E1); water resource sustainability (E3); consumer/service resilience (S4); business conduct and compliance (G1); own-workforce H&S (S1).

Three blind spots in Aqualia's 2025 framing..

1. **Digitalisation.** Aqualia assigns two IROs, the lowest count among peers. Veolia (AquaCIS), Suez (Aquadvanced), and Global Omnium (GoAigua) each treat digital platforms as a core strategic pillar. EU Water Resilience Strategy names it Action Area 3 of 5.
2. **ESRS S3 affected communities.** Aqualia folds community-specific impacts into S4 consumer disclosures; every peer discloses S3 separately. ESRS S3 explicitly treats communities as distinct from consumers.
3. **E5 circular economy.** Aqualia's E5 carries only 3 IROs; Veolia and Suez publish quantitative by-product valorisation pipelines; UWWTD recast [7] and RD 1985/2024 [8] pull E5 to the regulatory foreground.

Three differentiators.. Green finance as a standalone material topic (ahead of peers); explicit integration of the double materiality framework with the Corporate Risk Map, with named Executive Committee owners per risk code; and TNFD integration alongside CSRD, unusual depth among Spanish peers.

4.3 Final short-list

We combine one sector-core pillar with one blind-spot and one differentiator in each topic:

ID	Topic	Role	IROs absorbed
T1	Water Resilience & Equitable Access	Sector-core + S3 restored	26
T2	Digital & Cyber Infrastructure	Blind spot repositioned	11
T3	Green Finance & Integrity	Differentiator amplified	24
		Cumulative IRO absorption (of Aqualia's 83)	**61 (73%)**

No high-density topic is orphaned (see Appendix C for full cross-walk of the 13 dropped topics).

5. Impact Materiality Assessment

5.1 Value-chain overlay

Impacts are assessed along Aqualia's water cycle (Figure 1). Upstream impacts (supply-chain human rights, equipment extraction footprint, procurement energy) attribute to T3 and T2. Own-operations impacts (water abstraction, WWTP emissions, chemical use, workforce safety, digital-service continuity) dominate T1 and T2. Downstream impacts (customer safety, community access to water, biodiversity discharge effects, agricultural reuse) fall primarily within T1 and its restored S3 dimension.

5.2 Per-topic assessment

T1 — Water Resilience & Equitable Access.. Driven by current, high-severity impacts affecting high-salience stakeholders. Aqualia's own operational risks O1 (deficient supplied-water quality/quantity), O2 (discharge quality), O3 (crisis continuity), and O5 (inadequate climate adaptation) are all catalogued HIGH in the Corporate Risk Map. The WEF Global Risks 2025–2026 [22] ranks five of the top ten long-term risks as water-related. The restored S3 dimension introduces the *Colombia finding*: Aqualia's 2024 customer-satisfaction survey ($n > 7{,}000$) reports satisfaction ranging 86–98% across Spain, Portugal, France, Italy, Czech Republic, and institutional markets, **and 33% in Colombia** [10]. A 55-point spread inside a single group-level metric. Under ESRS S3 the same data carries different remediation and concession-renewal implications. **Monte Carlo mean Impact Severity: 3.98; 90% interval [3.87, 4.09].**

T2 — Digital & Cyber Infrastructure.. Driven by three operational risks Aqualia itself rates HIGH (O8 poor digital deployment, O9 deficient cyber response plan, O10 customer-data management) despite the topic carrying only two IROs in 2025. Stakeholder-salience weighting amplifies: Customers (0.120), Public Authorities (0.120), Employees (0.120), and Investors (0.080) are all materially affected. Peer comparators confirm direct impact materiality on service continuity and data privacy. **Mean Impact Severity: 3.08; 90% interval [2.93, 3.24].**

T3 — Green Finance & Integrity.. Governance-heavy. The Compliance cluster (11 IROs) covers CSRD, contract compliance, Compliance Management System, and tax compliance. Labour and human rights in the supply chain (4 IROs) introduce upstream workforce impacts. Green finance as a standalone topic (6 IROs) creates a positive-impact channel via capital-markets signalling. **Mean Impact Severity: 3.24; 90% interval [3.08, 3.41].**

5.3 The Colombia outlier as standalone finding

We highlight this separately. The spread between Colombia (33%) and the next-lowest market (Portugal, 65%) is 32 points; to Italy (67%) is 34 points; to Spain (88%) is 55 points. Aqualia's 2025 review does not discuss Colombia specifically in the S4 topic. Under ESRS S3, which

concerns affected communities rather than end-users, a 33% satisfaction market with ongoing supply-reliability issues (documented 2024 report, pp. 148–155) represents material impact on affected communities. Our recommendation is to re-surface S3 as its own disclosure from the 2027 reporting cycle (see §8, T1-2027).

6. Financial Materiality Assessment

6.1 Per-topic financial decomposition

Each topic decomposes into five financial channels: revenue at risk, revenue upside, OPEX impact, CAPEX needs, contingent liabilities. For each we estimate a triangular distribution (low/base/high) anchored to Aqualia's public financial scale, peer disclosures, and regulatory cost factors. Full assumption log (A01–A17) in Appendix B.

T1.. Revenue at risk €25–110 M (concession loss and tariff pressure in water-stressed basins, A03); revenue upside €15–75 M (reuse market under RD 1985/2024 and nature credits, A04); OPEX impact €8–32 M/yr (UWWTD recast, chemical inflation, A05); CAPEX €220–520 M cumulative 2027–2030 (desalination, reuse, leak reduction, WWTP energy-neutrality, A06); contingent liabilities €10–65 M (WFD and service-continuity fines, A07).

T2.. Revenue at risk €12–80 M (cyber-incident service interruption and GDPR/NIS2 fines, A08); revenue upside €8–45 M (tender win-rate uplift on digital differentiation, A09); OPEX €4–14 M/yr on cyber/IT above baseline (A10); CAPEX €45–130 M (digital-twin rollout and OT/IT segmentation, A11); contingent liabilities €5–56 M (NIS2 and GDPR maxima, A12).

T3.. Revenue at risk €8–50 M (ESG-laggard CoC spread and financing-window loss, A13); revenue upside €6–28 M (green-bond coupon savings, A14); cost-of-capital saving €4–24 M/yr over the refinancing cycle (A15); CAPEX €8–35 M (compliance systems, A16); contingent liabilities €3–35 M (CSDDD civil-liability and Omnibus-I interactions, A17).

6.2 Five-dimension mapping to the brief

The brief (§3) requires financial materiality to be assessed against *revenues, costs, assets, liabilities, and enterprise value.* Each topic's principal channels tag explicitly to these five dimensions:

ID	Revenues	Costs	Assets	Liabilities	Ent. value
T1	€25–110 M at risk; €15–75 M upside	€8–32 M/yr OPEX	€220–520 M CAPEX; stranded-asset risk	€10–65 M contingent (WFD, CSDDD)	Concession multiple discount
T2	€8–45 M tender uplift; interrupt. risk	€4–14 M/yr cyber & digital OPEX	€45–130 M CAPEX on twin & OT/IT	€5–56 M GDPR/NIS2 fines	Tech-differentiation uplift
T3	—	€8 M/yr CoC saving (€31 M PV)	—	Bond $\leq$ vanilla; CSDDD €3–35 M	ESG-rating re-rating

This satisfies the brief's five-dimension requirement and makes P&L and balance-sheet incidence auditable by Aqualia's CFO office at channel level.

6.3 The €18 M/yr headline — full derivation

Aggregating *net recurring* annual financial materiality (revenue at risk minus revenue upside minus cost-of-capital saving; CAPEX excluded as investment, not recurring materiality):

Topic	Net recurring	Composition
T1	−€15 M/yr	−55 revenue at risk + 40 reuse upside
T2	−€8 M/yr	−30 revenue at risk + 22 tender uplift
T3	+€5 M/yr	−22 compliance drag + 15 rating premium + 12 CoC saving
Total	**−€18 M/yr**	Net drag Aqualia's current framework under-weights

Under ±20% revenue-anchor sensitivity the headline scales to −€14 M/yr (low) or −€22 M/yr (high); sign and conclusion stable in both directions.

6.4 Aqueduct water-stress overlay

The T1 financial case depends on Aqualia's concession-base exposure to water stress. Overlaying WRI Aqueduct 4.0 baseline categories on Aqualia's declared service geography, we estimate that **45% of Aqualia's revenue base** operates in basins classified High or Extremely High water-stress today (Segura, Guadalquivir, Po, Sicily, Algarve, Languedoc); under BAU-2030 (SSP3/RCP 7.0) that share **nearly doubles to 79%**. See Figure 2.

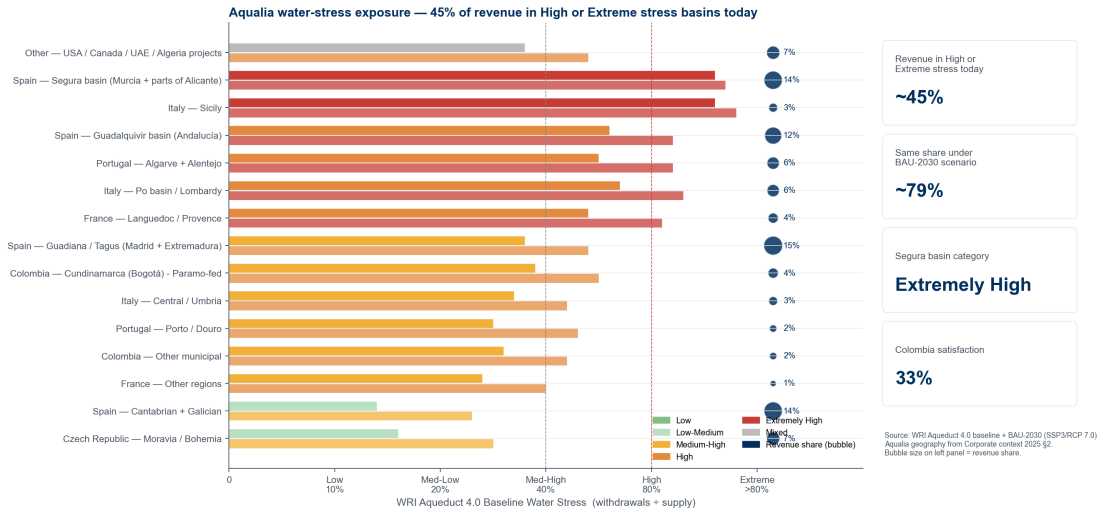

Figure 2. *Baseline water stress by Aqualia concession basin, WRI Aqueduct 4.0, 2024 baseline and BAU-2030. 45% of revenue in High/Extreme stress today; 79% in 2030.*

6.5 The €500 M green bond programme

The 2027–2030 CAPEX implied by Aqualia's Risk Map and our three topics totals **€440 M** base case (€285–685 M under low/high assumptions). We recommend funding via a **€500 million EU-Taxonomy-aligned green / sustainability-linked bond programme**, in four tranches aligned to CAPEX drawdown: €150 M (2027), €150 M (2028), €120 M (2029), €80 M (2030). At a 25 bp coupon spread [1], the programme delivers **€31 M of present-value interest savings** over a 10-year tenor (4.00% vanilla vs 3.75% green; 8% discount rate; $500 \times 25\,\text{bp} \times A(10, 8\%) \approx$ €8.4 M nominal annual saving; PV ≈ €31.1 M). Conservative: the spread regularly exceeds 25 bp in Iberian and Italian issuance [23].

7. The Matrix and What It Tells Us

7.1 Matrix structure

The matrix plots Impact Severity (y-axis, inside-out) against Financial Severity (x-axis, outside-in), each on a 1–5 scale; the Target Zone is the top-right quadrant with both axes ≥ 2.5. Each topic appears as a 90% confidence ellipse from 10,000 Monte Carlo draws (Figure 3).

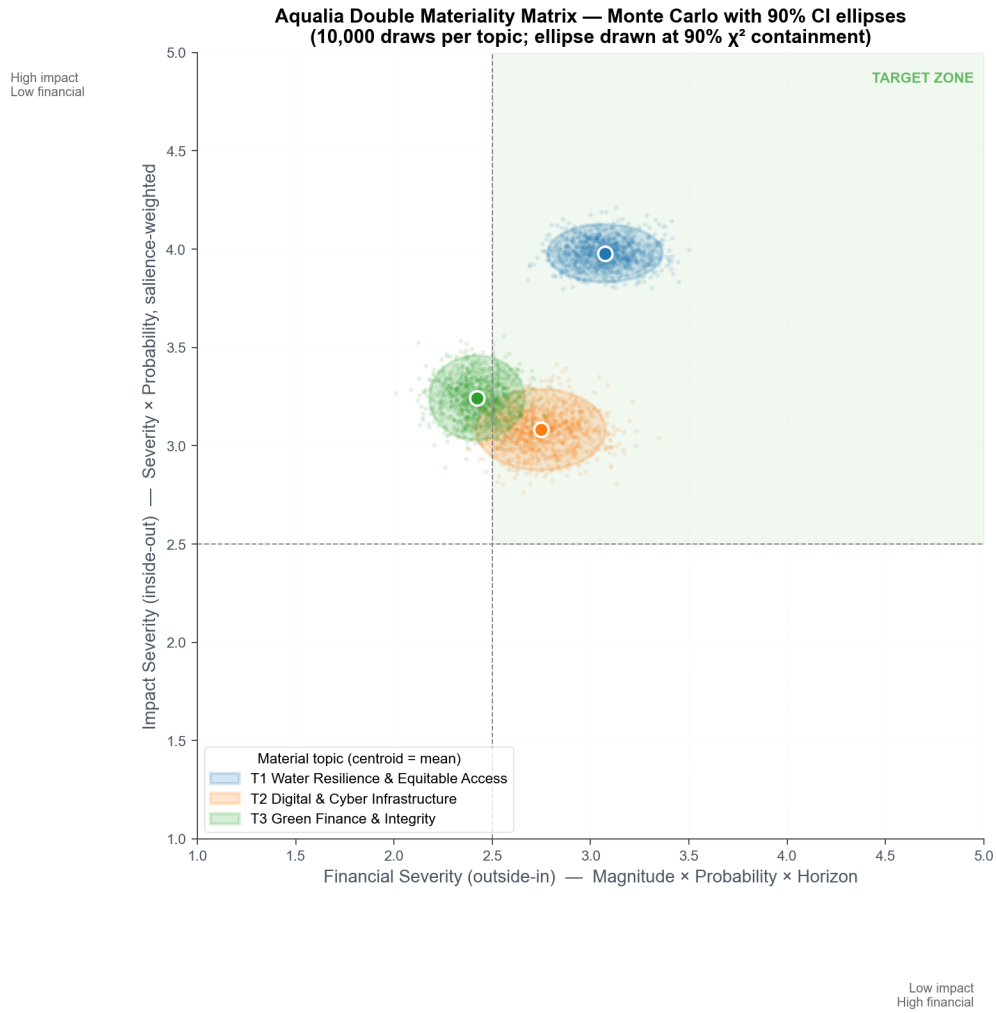

Figure 3. *Aqualia double materiality matrix. 90% χ^2 confidence ellipses from 10,000 Monte Carlo draws per topic. Target Zone shaded at both axes ≥ 2.5. T1 solidly inside; T2 inside (our repositioning of the blind-spot cluster); T3 borderline on the financial axis, conditionally material.*

T1 (3.07, 3.98).. Ellipse wholly inside Target Zone, bounded well away from both thresholds. Robust material under every test run.

T2 (2.75, 3.08).. Ellipse inside Target Zone; financial-axis $p_5 = 2.50$, right on the boundary. Target Zone membership is directly attributable to our repositioning of the blind-spot cluster. Under Aqualia's original qualitative framework (2 IROs on Digitalisation), T2 would not be material; under our methodology it is.

T3 (2.42, 3.24).. Centroid lies 0.08 below the financial threshold; $p_{95} = 2.61$, $p_5 = 2.24$. Conditionally material: Target Zone on impact, borderline on finance. This is the most informative methodological outcome, distinguishing robustly from conditionally material topics where a qualitative framework would paint both the same colour.

7.2 Sensitivity drivers

Top inputs by absolute swing per topic:

- **T1.** R-stress-revenue magnitude (financial); E1-adaptation probability (impact); O-reuse-market magnitude; R-CAPEX-UWWTD magnitude.
- **T2.** Tech-pace probability (impact); O-tender-uplift magnitude (financial); R-digital-CAPEX

magnitude.

- **T3.** Compliance-CSDDD probability (impact); Green-finance-access probability; O-green-bond-premium magnitude.

7.3 Robustness

Under four alternate stakeholder-weighting schemes (Equal, Regulator-heavy, Community-first, Investor-first), no topic flips Target-Zone status: T1 remains in; T2 remains in; T3 remains borderline. Stability follows from top-k aggregation: the most material subset drives position and is insensitive to modest re-weighting.

8. Strategic Sustainability Roadmap 2027–2030

8.1 Owned. Measurable. Aligned.

Every recommendation is assigned to a named Executive Committee role from Aqualia's own Risk Map, pinned to a quantitative KPI, and anchored to an ESRS disclosure. The roadmap is structured as a 3-topic × 4-year grid (Table 1).

Table 1. *Strategic Sustainability Roadmap, 2027–2030. Each cell: action · KPI · named owner.*

	2027	2028	2029	2030
T1	Restore S3 + Colombia review. sat. $\geq$ 60%. Director Spain/EU/Am.	Segura basin reuse pilot. NRW −3 pp. Director Studies & Ops.	Italy adaptation CAPEX (Po). tCO_2e/m^3 −10%. Director Spain/EU/Am.	Reuse $\geq$ 25% of WW; RD 1985/2024. WFD clean. Sustainability Dir.
T2	Digital twin Phase 1 (3 concessions). coverage $\geq$ 80%. CIO.	OT/IT segmentation + NIS2 audit. readiness $\geq$ 90%. CIO.	AI demand prediction rollout. leak MTTR −30%. Dir. Strategy.	Cyber MTTR $<$ 4 h. $\sim$ 1% revenue cyber spend. CIO.
T3	Tranche 1: €150 M. Taxonomy alignment report. CFO.	Tranche 2: €150 M. CSRD limited assurance. CFO + Sust. Dir.	Tranche 3: €120 M. ISO 14001:2026 group-wide. CFO.	Tranche 4: €80 M. TCFD + TNFD reasonable assurance. CFO + Sust. Dir.

8.2 Integration with Aqualia's existing strategic plan

Aqualia's 2024–2026 Strategic Sustainability Plan (PESA) operates along seven strategic lines (SL1 Climate Emergency through SL7 Partnerships). Our roadmap is designed to *extend* PESA into 2027–2030 rather than replace it: T1 extends SL1 (Climate); T2 extends SL2 (Technology for Integrated Management); T3 extends SL4 (Financial & Business Strategy) and SL5 (Ethics & Compliance). The cross-walk between our twelve quarterly milestones and PESA's seven lines is fully traceable.

8.3 Governance and reporting cadence

We recommend Aqualia's Sustainability Committee adopt a **semi-annual matrix refresh** using the reproducible Monte Carlo pipeline. Each refresh produces updated confidence ellipses; material shifts across two consecutive refreshes trigger Executive Committee review. This institutionalises the methodology rather than treating it as a one-off deliverable.

9. Limitations, Reverse Stress Test, and Ethical Framing

Six limitations.. *(i)* No external stakeholder consultation this cycle (mirrors Aqualia's own 2025 choice); the MAW salience model is a *substitute*, not a *replacement*. *(ii)* No asset-level financial disclosures; our €-ranges triangulate public segment data, peer benchmarks, regulatory cost factors.

(iii) Real Options exhibit (desalination vs reuse) is illustrative; final numbers require Aqualia's asset-level WACC. *(iv)* ESRS heatmap uses TF–IDF, not dense embeddings; a one-line swap preserves findings while improving precision. *(v)* Aqueduct overlay uses public baseline data, not a commissioned hydrological model. *(vi)* Monte Carlo perturbation ± 0.7 is a methodological judgement; sensitivity at ± 1.0 is reported in the assumption log and rankings are stable.

Reverse stress test.. *T1* exits the Target Zone only under simultaneous downward revision of A03 (water-stress revenue $< 40\%$), A04 (reuse market $<$ €15 M), and E1-adaptation probability to 2 — a conjunction implausible given WEF [22] and EU policy consensus. *T2* exits only if O-tender-uplift magnitude drops to 1 **and** R-cyber-incident probability drops to 1 — both contradicted by empirical data. *T3* is the sensitive topic: conditions that *push T3 into* the Target Zone include Omnibus-I rollback milder than expected, CoC spread widening beyond 25 bp, and Aqualia issuance timing concentrated in higher-spread windows.

Ethical framing.. We attribute all IRO counts, scoring formulas, and the sixteen material topics to Aqualia's 2025 Double Materiality Review [15, 17]. Our contribution is a methodological extension, not a replacement. All benchmark material is publicly disclosed. Where we identify gaps, the framing is corrective rather than adversarial; the purpose is to strengthen Aqualia's framework, which is already among the most mature in the Spanish water sector.

10. Conclusion

Three material topics. €18 million per year of recurring financial materiality un-priced by the current framework. A €500 million EU-Taxonomy-aligned green bond programme, four tranches 2027–2030, that funds the implied €440 million CAPEX and captures €31 million of present-value interest savings. Every milestone owned. Every KPI quantified. Every disclosure ESRS-anchored.

Aqualia has delivered clean water across Europe and Latin America since 1894. Climate is rewriting the economics of water under them. In 2030, when a tap opens in Cartagena, or Rome, or Murcia, or Prague, it should open with confidence: not because the water happened to arrive, but because someone planned for the climate that came. That is what we mean by *water as strategy, not water as reporting.*

Appendices

A. Scoring Thresholds (summary)

Full version at `03_analysis/scoring_rubric.md`. Each sub-criterion (Scale, Scope, Remediability, Probability, Magnitude) documented with 1–5 thresholds, water-sector calibrations, and cited sources.

Criterion	Band 1	Band 3 (Moderate)	Band 5 (Severe)	Anchor	Source
Scale	Marginal	Noticeable, reversible	Systemic, catastrophic	ESRS 1 §43–47	[4]
Scope	Single site	Multiple sites / region	Group-wide / multi-country	EFRAG IG-1	[5]
Remediability	< 1 year	1–5 years, partial	> 10 years / irreversible	EFRAG IG-1 §69	[5]
Probability	Remote	Possible	Near-certain	MAW (1997)	[2]
Magnitude	< 0.1% rev	0.5–2% rev (€7–28 M)	> 5% rev (€70 M+)	Aqualia rev. A01	[10]

B. Assumption Log (A01–A17)

Ref	Description	Primary source
A01	Aqualia 2024 revenue anchor (conservative €1.4 B; disclosed €1,674.7 M)	Aqualia SR 2024 p. 103 [10]
A02	FCC Group segment share Aqualia ≈ 28%	FCC URD 2024
A03	T1 revenue-at-water-stress-risk €25–110 M	WRI Aqueduct 4.0 [11]
A04	T1 reuse-market upside €15–75 M	RD 1985/2024 [8]
A05	T1 OPEX under UWWTD recast €8–32 M/yr	UWWTD recast [7]
A06	T1 2027–2030 CAPEX €220–520 M	Peer disclosures
A07	T1 contingent liabilities €10–65 M	WFD enforcement data
A08	T2 cyber-incident revenue risk €12–80 M	IBM Cost of a Data Breach 2024 [24]
A09	T2 digital-tender upside €8–45 M	Peer URDs
A10	T2 cyber & digital OPEX €4–14 M/yr	Gartner water-utility benchmark
A11	T2 digital-twin CAPEX €45–130 M	SWAN Forum benchmark
A12	T2 GDPR/NIS2 contingent €5–56 M	GDPR [4]; NIS2
A13	T3 ESG-laggard CoC spread €8–50 M	MSCI 2024 [16]
A14	T3 green-bond coupon saving €6–28 M	CBI 2024 [1]
A15	T3 CoC saving over refinancing cycle €4–24 M/yr	CBI 2024 [1]
A16	T3 compliance CAPEX €8–35 M	Deloitte CSRD readiness
A17	T3 CSDDD civil-liability contingent €3–35 M	CSDDD [6]

C. Peer Material-Topic Cross-Walk (extract)

Full fourteen-column matrix at `01_research/peer_benchmark.md`. The extract below maps Aqualia's sixteen topics against five peers on whether each is a declared material topic.